Materials at Imperial

Overview

DEGREE	DURATION	DEPARTMENT
MEng / BEng	**3-4 years**	**Materials**

MINIMUM ENTRY STANDARD	ADMISSIONS TEST	INTERVIEW
AAA (IB 38)	**None required**	**Yes (In-person for home students/online for internationals)**

APPLICATIONS:PLACES		
8:1		

The Materials Science and Engineering degree at Imperial College London combines principles from physics, chemistry, and engineering to understand how materials are designed, processed, and used in modern technologies. Covering topics from metals and polymers to biomaterials and nanotechnology, the programme develops both theoretical knowledge and practical skills through laboratory work and research-led teaching. Graduates gain the expertise to innovate across industries including energy, healthcare, aerospace, and advanced manufacturing.

A LEVELS

Required: A in Maths, A in Chemistry OR Physics

Recommended: Maths, Chemistry, Physics, Further Maths

HOW TO APPLY

Personal Statement

Question 1: Why do you want to study this course or subject?

In this question you want to show a genuine interest in the course. You don’t necessarily need to have experience, but showing initiative and passion is really important. This can be by exploring a topic related to materials science that interests you, a book you’ve read, or even a personal experience which motivated you to pursue this degree. The more personal you make it, the better! The department wants students who are genuinely passionate and want to hear your motivations for wanting to study materials science. This is your chance to show them what drives you and make them interested in your application.

Question 2: How have your qualifications and studies helped you to prepare for this course or subject?

This question is all about exploring how your current A-levels and/or past studies have prepared you to study this course at university. How did doing A-level maths prepare you for the engineering side of this degree? Was there a particular topic in Chemistry/Physics that you particularly enjoyed? What have your other subjects taught you that will help you in this degree?

Question 3: What else have you done to prepare outside of education, and why are these experiences useful?

This is where you show how motivated you truly are. Any short courses you’ve done, part-time jobs, or work experiences which have helped you gain the necessary skills to thrive at university and during this degree. These experiences dont necessarily need to be related to materials but should have a

Interview Prep:

Our Services:

THE COURSE

The department offers 4 courses, all of which have the same teaching for the first 2 years. The courses are as follows:

- Materials Science & Engineering (BEng)
- Materials Science & Engineering (MEng)
- Materials with Nuclear Engineering (MEng)
- Biomaterials with TissueEngineering (MEng)

YEAR 1:

Core modules

- Mathematics and Computing 1
- Performance of Structural Materials
- Engineering Practice 1
- Materials Processing 1
- Structure 1
- Properties 1

YEAR 2:

Core Modules

- Mathematics and Computing 2
- Engineering Practice 2
- Materials Characterisation
- Properties 2
- Structure 2
- Performance of Functional Materials

YEAR 3:

Core modules

- Materials Processing 2
- Processing Laboratory
- Theory and Simulation of Materials
- Managerial Economics
- Research Techniques
- I-Explore

Optional modules

- Sustainable Metallurgy
- Powder-Based Manufacturing
- Nanomaterials
- Math and Quantum Mechanics
- Biomaterials
- Optoelectronic Materials
- Advanced Characterisation
- Nuclear Fusion
- Magnetic Materials & Devices
- Materials Sustainability

In Year 3, you will be given the opportunity to choose 4 optional modules to take alongside the core modules. Those on the Nuclear/Biomaterials courses may have some optional modules already selected for them. I-Explore is also compulsory in this year, where you will have the chance to explore a non-STEM field in the form of an extra module. These credits will count towards your degree.

YEAR 4:

Core modules

- Individual Project

Optional modules

- Nuclear Materials
- Materials for Energy Conversion and Storage
- Density-Functional Theory
- Ceramics and Composites for Extreme Environments
- Materials for Quantum and Semiconductor Technologies
- Machine Learning for Materials
- Tissue Engineering
- Nanomedicine
- Aerospace Metallurgy
- Microstructure Engineering in Metal Processing

The 4th year is an integrated masters year in which you will undertake an individual masters project alongside 4 optional modules.

STUDENT LIFE